

Proposition 10.3 (Multifold binomial sums of powers negated). *For non-negative integers n, m , and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m (-1)^{m+k} \left\{ \binom{n+t+\frac{k}{2}+r}{k+r} + \binom{n+t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s+t+\frac{k}{2}}{k+r-s} + \binom{r-1-s+t+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k t^m. \end{aligned}$$

Proof. By setting $t \rightarrow -t$ into Proposition (10.1), and by $\delta^k(-t)^m = (-1)^{m+k} \delta^k t^m$. □